



Assessment of Sustainable Supply Chain Management for Mangrove Crab on Bangka Island



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Abstract: Mangrove crab (*Scylla serrata*) represents one of the most economically valuable fisheries commodities in Indonesia. However, the long-term sustainability of mangrove crab production on Bangka Island, Bangka Belitung Islands Province, has been threatened by the declining availability of wild seed resources, primarily as a consequence of mangrove ecosystem degradation, land-use conversion for industrial development, and expanding residential areas. Therefore, the sustainability performance of the mangrove crab supply chain was evaluated through a sustainable supply chain management framework. A mixed-methods approach was employed, in which qualitative analyses were conducted to characterize the supply chain structure, while quantitative assessments were performed using multi-aspect sustainability analysis supported by Exsimpro. The results demonstrated the existence of four distinct channels, each exhibiting different levels of operational efficiency. Sustainability performance across the supply chain was found to be relatively low. Nevertheless, substantial opportunities for improvement were identified. The findings highlight the urgent need for integrated sustainable supply chain management implementation to balance economic benefits with ecosystem conservation and resource sustainability. These results provide a scientific basis for policymakers, fisheries managers, and supply chain actors to develop targeted interventions that support the sustainable utilization of mangrove crab resources while enhancing the resilience and competitiveness of the fisheries sector on Bangka Island.

Keywords: Mangrove crab; Sustainable supply chain management; Bangka Island; Multi-aspect sustainability analysis; Fisheries sustainability

1. Introduction

The Bangka Belitung Islands Province is one of the archipelago provinces in Indonesia. Bangka Belitung is divided into two archipelago areas: Bangka Island and Belitung Island. This province has a land area of 16,281 km² and a water area of 65,301 km². Almost 80% of the area of Bangka Belitung is water. The vastness of this water area affects the potential of existing resources. The strong fisheries sector focuses on people's lives. There are many fishery commodities that can be developed, among which mangrove crab (*Scylla serrata*) is particularly important (Falatehan et al., 2025). In Indonesia, the Ministry of Marine Affairs and Fisheries has inaugurated crab villages in the Bangka Island area, especially Central Bangka in 2023. This inauguration is not just a formality, but a strategy to improve the economic quality of the community through the appointment of regional potential. After Central Bangka, other areas, especially Pangkalpinang, West Bangka, Bangka Induk and South Bangka, hereinafter referred to as Bangka Island, have participated in increasing mangrove crab production. This causes Bangka Island to become one of the islands known as the largest producer of mangrove crabs in Bangka Belitung.

Basically, the habitat of mangrove crabs is mangrove forests or mangroves. The mangrove forest on Bangka Island has contributed substantially to the high production potential of mangrove crabs. The total mangrove forest area on Bangka Island, encompassing the five previously mentioned districts and municipalities, is estimated at 58,578.65 hectares. The mangrove forest area consists of 11.68 hectares (0.01%) in the Pangkalpinang area, about 4,470.89 hectares (5%) in the Central Bangka area, about 42,529.08 hectares (44%) in the West Bangka area, about 38,957.14 hectares (40%) in the Bangka Induk area, and 11,567 hectares (12%) in the South Bangka area.

Based on the number of mangrove forests on Bangka Island, around 6.4%, or around 6,201.25 hectares, has been reported to be damaged. This disturbs the natural habitat of mangrove crabs, resulting in scarcity and price increases in the market. Therefore, preventive measures should be taken. One alternative that can be made is to improve the efficiency of the mangrove crab supply chain through the establishment of sustainable supply chain management.

Supply chain management is a management system that has been widely analyzed for more than two decades because its effectiveness and risks have evolved with technological advances (Dubey et al., 2021; Ivanov et al., 2019; Maheshwari et al., 2021). According to Wisner et al. (2008), supply chain management requires “attention to where materials come from, how their suppliers’ products and services are designed, produced and transported, and how their own products and services are produced and distributed to customers.” Based on this sentence, it can be understood that in the supply chain management process, it is necessary to pay attention to where the product comes from, how the product is designed, and how the distributor delivers the product to retailers or even end consumers. Effective supply chain management can enhance customer service performance by facilitating the systematic coordination of raw materials supply chain stakeholders (Govindan et al., 2020). A supply chain scheme that reflects sustainability is a concept of sustainable supply chain management (Bagherpasandi et al., 2024). Sustainable supply chain management can be defined as a supply chain system that pays attention to the environment, economy, and society in its implementation (Srivastava et al., 2022; Zimon et al., 2020). Each company that implements sustainable supply chain management should carry out the main activities, namely sustainable supplier management, selection of raw material sources, corporate and social responsibility, and life cycle assessment (Barros et al., 2021; Burke et al., 2023; Grolleau et al., 2023). The sustainable supply chain management scheme is closely related to the 2030 Sustainable Development Goals (SDGs) (van Zanten & van Tulder, 2021) because its implementation integrates environmental, social, and economic sustainability considerations throughout the supply chain. In this context, the goals can be categorized into three dimensions: the social dimension (SDGs 1, 3, 4, 5, and 10), the economic dimension (SDGs 8, 9, 11, and 12), and the environmental dimension (SDGs 13, 14, and 15) (Körffgen et al., 2018).

Research related to sustainable supply chain management has been carried out in several countries. In the United States, Wren (2022) discussed the environmental impact of the supply chain of fast fashion brands, focusing on the generation of textile waste and its contribution to climate change. In Nigeria, Abaku & Odimarha (2024) discussed the influence of the medical industry. To create sustainable supply chain management, it was found that fewer environmentally unfriendly materials should be used, and should be replaced with blockchain, the Internet of Things, and artificial intelligence technologies.

The application of sustainable supply chain management in this study focuses on a different industry, namely the fisheries industry. The fisheries sector in question is mangrove crab which is a commodity with economic value. Its sustainable supply chain management is assessed through a different method, which involves an assessment of supply chain sustainability that focuses on three aspects of the SDGs (economic, social, and environmental). The assessment criteria are determined based on the sustainable supply chain management matrix created by Zimon et al. (2020). The data obtained is processed using Exsimpro software with the multi-aspect sustainability analysis approach. This study aims to determine the sustainability status of the mangrove crab supply chain on Bangka Island. The more sustainable the existing supply chain management system, the more balanced it will be between consumer demand and the current mangrove crab production.

2. Methodology

This research was conducted in five cities in Bangka Belitung Islands Province, including Bangka Tengah, Bangka Induk, Bangka Selatan, Bangka Barat, and Pangkalpinang. The respondents comprised mangrove crab supply chain actors who had been supported by the Regional Development Planning Agency, including 20 fishermen, 3 cultivators, 2 mangrove crab farmers, and 5 retailers. The research location was chosen deliberately because Bangka Island has a very high potential for mangrove crabs due to its large mangrove forest land. The mangrove crab cultivation sector was chosen as the focal business unit due to its central role within the mangrove crab supply chain. The research was carried out from July to September 2024.

As for the data collection process, the purposive sampling method was used. The method is deliberate selection based on predetermined criteria and snowball sampling, which is a sample selection that involves information from the previous sample to find samples that are difficult to reach. The core sample in this study was taken from mangrove crab business units on Bangka Island, such as cultivators, retailers, fishermen, and mangrove crab collectors. Two key informants were purposively selected from the Fisheries Service and the Regional Development Planning Agency of the Bangka Belitung Islands Province. A multi-aspect sustainability analysis method was used for the sustainable supply chain management analysis, aiming to find the sustainable status of the mangrove crab supply chain on Bangka Island and the strategies that must be carried out in the future. The analysis process using multi-aspect sustainability analysis was carried out through the Exsimpro software. Several things should be considered in the software, namely aspects and factors, indicators, questionnaires, and

respondents.

Aspects and factors are used as the sustainability parameters in this study. Although the aspects are very diverse, sustainability measurements are based on economic, social, and environmental aspects in this study. Meanwhile, the factors used are questions in accordance with the aspects and are based on sustainable supply chain management (recycling management system). In addition, questionnaires are a feature in Exsimpro multi-aspect sustainability analysis that can be used to collect data from respondents as research objects.

In this study, two results can be interpreted as an assessment of the sustainability of the supply chain or mangrove crab sustainable supply chain management, including the status index and leverage factor. The status index is divided into aggregate statuses derived from the status of aspects (economic, social, and environmental) and possible future values (future conditions). To assess the sustainability status of the mangrove crab supply chain, the mathematical formulations and assessment attributes proposed by Bui et al. (2021) were adopted, as presented below.

2.1 Aggregate Status

The value of aggregate status is the average value of the performance status of economic, social, and environmental aspects. The aggregate value can be produced directly by calculating the average value of the aspect state, as per Eq. (1).

$$Y = (y_1 + y_2 + y_3 + y_4 + \dots + y_n)/n = (\sum y_n)/n \quad (1)$$

where, Y indicates the status value (sustainability/performance), y denotes the aspect status value, and n indicates the number of aspects.

2.2 Aspect Status

The aspect status is visualized on the X -axis in the ordination graph. This value is calculated based on the mode of indicator assessment and is then divided by the best indicator value (good) for each factor. Finally, it is averaged from all factors (Tirkolaee et al., 2021).

$$y = (y_{f1} + y_{f2} + y_{f3} + y_{f4} + \dots + y_{fn})/fn \times 100\% = (\sum fn)/fn \quad (2)$$

$$y_{fn} = (Mo, fn)/Gfn \quad (3)$$

where, y_f denotes the factor aspect, Mo indicates the mode value, G denotes the highest score (good), and f indicates the factor value.

The G value reflects the maximum or minimum score obtained from the factors associated with assessed indicators. While Mo represents the most commonly occurring score based on the expert respondent's input.

2.3 Future Condition

This possible future value (future condition) describes the potential for changes in these factors in the future, whether there will be a decrease or perhaps an increase. In calculating the value of future conditions that will be visually depicted, the Y -axis can be used. The value of factor conditions in the future is calculated as follows.

$$Fc = (MoC_1 + MoC_2 + MoC_3 + MoC_4 + \dots + MoC_n)/n \times 25 \quad (4)$$

where, Fc indicates the future condition and C indicates the value of future conditions on the factor.

2.4 Leverage Factor

Leverage factors or driving factors reflect the elements that have the greatest influence on changes in status, both in individual and overall aspects (Cristini et al., 2021). The value of the driving factor is calculated from the highest value produced by the sum between the maximum sensitivity and the actual sensitivity (Ibrahim & Sapanli, 2024). In addition, there are several things that affect the value of the factor, namely random iteration and uncertainty error. Mathematically, it can be seen in the equations as follows:

$$L = SM + SV \quad (5)$$

$$SM = 1/Gfn \quad (6)$$

$$SV = (Gfn - Mofn) \times SM \quad (7)$$

where, L indicates the leverage factor value, SM denotes the maximum sensitivity, and SV indicates the sensitivity value.

2.5 Random Iteration

In the multi-aspect sustainability analysis, validation is carried out by calculating random values from opinions or modes in factor assessment (Raj et al., 2022). This process aims to evaluate the state of the factor based on random values and measure the deviation from the mode. This assessment uses the opinions of experts or respondents. The equations used in the validation procedure are presented below.

$$P = fr/(\sum ln) \quad (8)$$

$$K_n = P_{((n+(n+1)))} \quad (9)$$

$$l_r = K_n \times 100 \quad (10)$$

$$A_{rn} = Rln(1,100) \quad (11)$$

$$Si_f = \sum if(Arn;ln) \quad (12)$$

$$Avg Si_f = Si_f/Rln \quad (13)$$

where, P indicates the probability, l_r denotes the random interval, ln indicates the indicator value (with a minimum score of 0), fr denotes the frequency of indicator values, K indicates the cumulative value, Rl denotes the random interaction, Si_f indicates the simulated indicator values, and $Avg Si_f$ denotes the random simulation average.

2.6 Sustainability Attributes

The sustainability attributes developed in accordance with the multi-aspect sustainability analysis methodology can be used to assess the sustainability of the supply chain. The attributes are classified into three sustainability dimensions: economic, social, and environmental. Table 1 shows the attributes.

Table 1. Sustainability assessment attributes used in the multi-aspect sustainability analysis

Variable	Attribute	Description
Economic Aspect		
<i>NP</i>	Net profit	Net profit earned by respondents after deducting costs. The Bangka Belitung minimum wage is IDR 3,640,000.
<i>MCT</i>	Mangrove crab sales target	The sales target refers to the number of marketing institutions involved in marketing mangrove crab products.
<i>MCP</i>	Mangrove crab production/harvesting capacity	Mangrove crab production capacity refers to the catch volume of fishers, cultured production of farmers, and purchase volume of collectors and retailers.
<i>LI</i>	Labor income	Income earned by workers, including family labor.
<i>PQ</i>	Product quality	Product quality is determined by weight, which is classified into three crab grades.
<i>MA</i>	Market access (ease of selling mangrove crabs)	Market access refers to the ease with which respondents can sell mangrove crabs across different market ranges, from local and domestic markets to international markets.
<i>TD</i>	Technology development (e.g., fishing gear and production equipment)	Technology development refers to the use of new inputs that support respondent productivity.
Social Aspect		
<i>LEA</i>	Local employment absorption	Local employment absorption means that workers employed by respondents come from the surrounding community rather than from outside Bangka Island.
<i>WFC</i>	Workforce creativity	Creativity refers to the ability to support respondent productivity.
<i>EAH</i>	Ease of access to healthcare	Healthcare access includes community health centers, village health posts, pharmacies, and hospitals.
<i>USP</i>	Use of occupational safety equipment/protective gear	Safety equipment includes items such as gloves to prevent crab pinches.

<i>WPC</i>	Workplace conditions	The conditions of the place of work: no wild animals and difficult tracks (for fishermen); good access to water and salinity and not polluted or with other problems (for cultivators); and a place where crab buying and selling transactions occur (for collectors and retailers).
<i>WFT</i>	Workforce training	Training includes harvesting, processing, and marketing activities.
Environmental Aspect		
<i>REC</i>	Reduction in energy consumption	Energy use refers to energy sources such as fossil fuels.
<i>UEM</i>	Use of environmentally friendly materials	Environmentally friendly materials are materials that do not pollute the environment. Examples include wood, bamboo, and other biodegradable materials that do not contaminate water.
<i>PEC</i>	Possession of environmental certification (from a credible institution)	Examples include Cradle to Cradle, Intertek, EcoVadis, ISO 14001, and other green industry certifications.
<i>WMG</i>	Waste management (crab shells)	The waste in this indicator is crab shells.
<i>UIL</i>	Use of business inputs from local suppliers	Inputs include vessels, equipment, processing tools, distribution tools, and marketing facilities.

Note: IDR = Indonesian rupiah.

3. Results

3.1 Supply Chain Overview in Bangka Island

Supply chains are generally a series of activities of actors involved in a marketing system. Mangrove crab commodities also have actors involved so that they form a supply chain. Each of these actors has a relationship with each other with the aim of providing mutual benefits for all parties.

Mangrove crabs on Bangka Island are also sold locally in the Bangka Belitung area. The available supplies are widely provided by fishermen or crab catchers in mangrove forest areas. Mangrove crab cultivation is still very rarely done by the community, because mangrove crab habitat engineering is very complex if made artificially. Mangrove crabs must lay their eggs in salt water and forage and rear in estuary water. This life cycle makes artificial habitat engineering technically challenging and economically inefficient, with only two cultivators being identified in the study area, located in Central Bangka and Pangkalpinang. The cultivation carried out is only the fattening process of 100-gram crabs caught from their natural habitat using crab box media or crab apartments. Figure 1 shows the differences in the volume of mangrove crab production from fishermen's catches and cultivation.

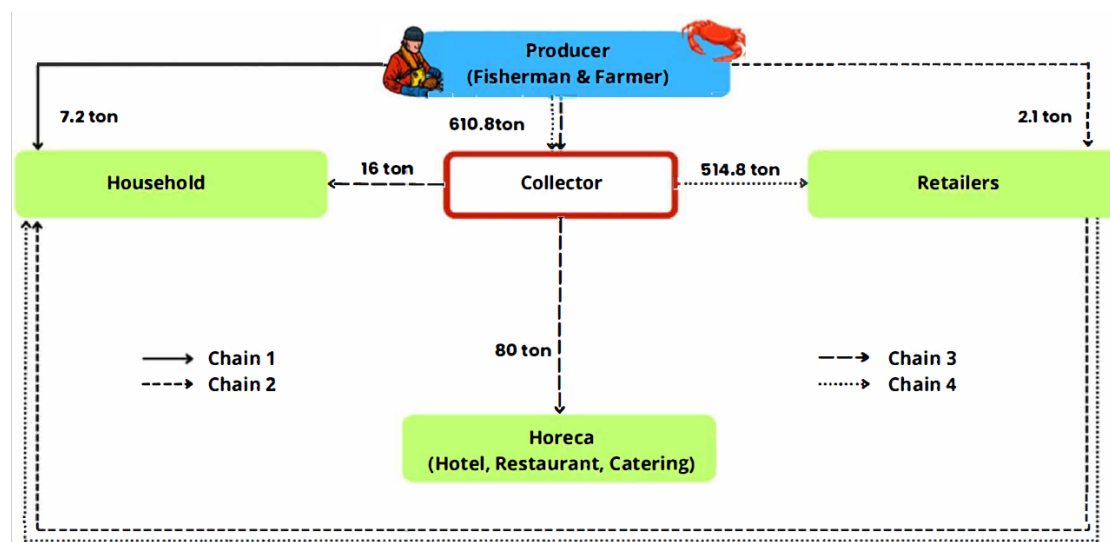


Figure 1. Mangrove crab supply chain on Bangka Island

Most of the mangrove crab demand in the Bangka Belitung is met through local production supplied by fishermen and cultivators. Crabs are distributed through collectors and marketers or retailers before reaching hotels, restaurants, and catering (horeca) and some crabs are sold directly from producers to households. Therefore, it can be said that the suppliers of mangrove crabs on Bangka Island consist of producers (fishermen and cultivators), collectors, and retailers.

Several actors in the mangrove crab marketing transact with each other to create a marketing pattern in the form of a mangrove crab marketing supply chain. The supply chain actors are engaged in production, distribution, and

marketing activities, ranging from the procurement of raw materials to the delivery of products to consumers. Some crab supply chains are as follows:

Chain 1: Producer → End Consumer

Chain 2: Producer → Retailers → End Consumer

Chain 3: Producer → Collector → End Consumer

Chain 4: Producer → Collector → Marketers or retailers → End Consumer

3.2 Sustainable Supply Chain Management Assessment

The sustainability analysis of the supply chain on Bangka Island was carried out using the multi-aspect sustainability analysis method with the help of Exsimpro software. Three aspects, namely economic, social, and environmental, were used. The sustainability assessment was divided into four statuses, including unsustainable, low sustainable, sustainable, and very sustainable. Meanwhile, the status of future performance consisted of greatly decreasing, decreasing, increasing, and greatly improving. Table 2 shows the details of the sustainability criteria and future performance.

Table 2. Classification intervals for sustainability status and future performance

Value	Sustainable Status	Future Performance Status
0–25	Unsustainable	Greatly decreasing
>25–50	Low sustainable	Decreasing
>50–75	Sustainable	Increasing
>75–100	Very sustainable	Greatly improving

In the sustainability analysis of the mangrove crab supply chain, economic, social, and environmental aspects were used. Further explanations related to sustainability aspects were the results of the status index, leverage factor, and random iteration.

3.2.1 Status index

A. Sustainability in the economic aspect

The sustainability of this economic aspect includes the attributes of net profit, sales target, production capacity, labor income, production quality, market access, and technology development. The results are shown in Figure 2.

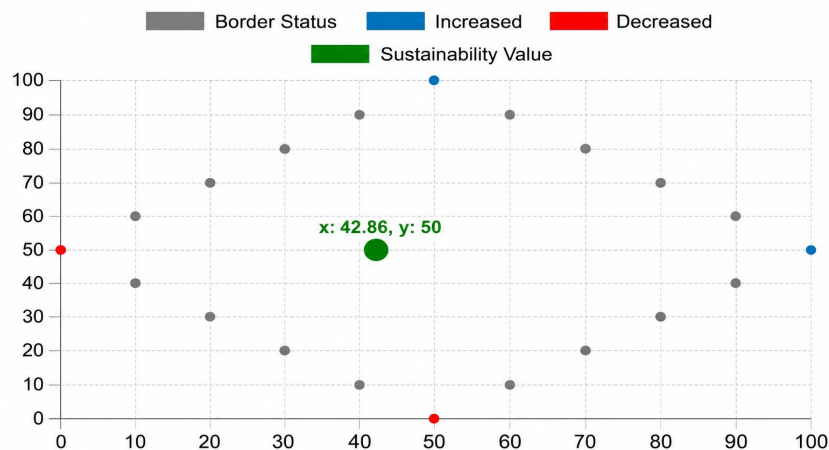


Figure 2. Sustainability status of the economic aspect

Based on Figure 2, the sustainability value of the economic aspect is 42.86, which is indicated by the X-axis on the ordination diagram. This value illustrates that the sustainability of the economic aspect is still in the low sustainable category. The value of possible future performance is 50, which is indicated by the Y-axis on the ordination diagram. This value illustrates that future performance is likely to improve.

B. Sustainability in the social aspect

The sustainability of the social aspect includes the attributes of labor absorption, workforce creativity, access to health, work safety costumes, workplace conditions, and skills training. The results are shown in Figure 3 below.

Based on Figure 3, the sustainability value of the social aspect is 30, which is indicated by the X-axis on the ordination diagram. This value illustrates that social sustainability is still in the low sustainable category. The

value of possible future performance is 50, which is indicated by the Y-axis on the ordination diagram. This value illustrates that future performance is likely to improve.

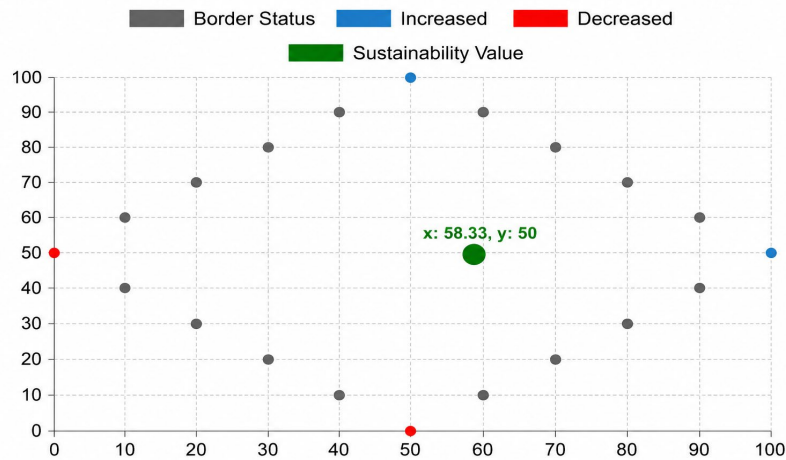


Figure 3. Sustainability status of the social aspect

C. Sustainability in the environmental aspect

The sustainability of the environmental aspect includes the attributes of the reduction of the amount of energy, the use of environmentally friendly materials, environmentally friendly certificates, waste management, and the use of local business inputs. The results are shown in Figure 4 below.

Based on Figure 4, the sustainability value of the environmental aspect is 30, which is indicated by the X-axis on the ordination diagram. This value illustrates that environmental sustainability is still in the low sustainable category. The value of possible future performance is 50, which is indicated by the Y-axis on the ordination diagram. This value illustrates that future performance is likely to improve.

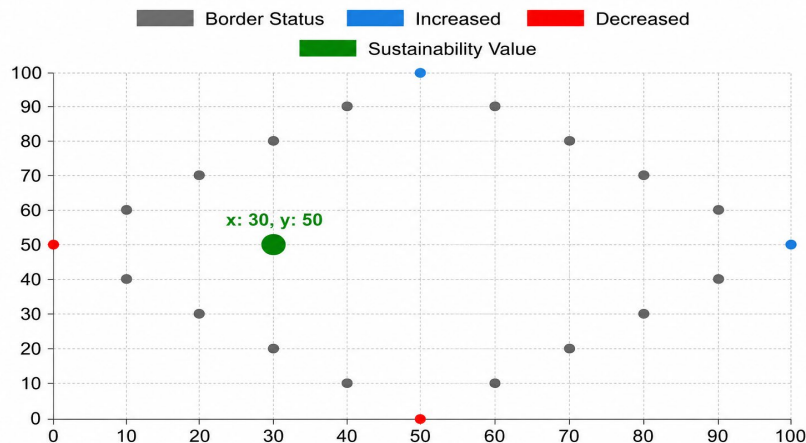


Figure 4. Sustainability status of the environmental aspect

D. Sustainability in aggregate

After the analysis of each aspect was carried out, all the sustainability values of economic, social, and environmental aspects were compared and an average of the sustainability values of the three aspects was calculated. Therefore, whether the existing supply chain is sustainable or not was assessed. Sustainability was analyzed from all existing supply chain actors, namely fishermen, cultivators, collectors, and retailers of mangrove crabs on Bangka Island. Figure 5 and Table 3 show the details of the overall sustainability value.

The results of the analysis show that overall sustainability is in the low sustainable category with an average score of 43.73. The social aspect occupies the highest position with a score of 58.33, which reflects relatively good social management, such as community participation and social welfare. On the other hand, the economic aspect has a score of 42.86, indicating that efficiency and economic management still need improvement. Meanwhile, the environmental aspect shows the lowest performance with a score of 30, indicating significant pressure on environmental sustainability, such as a lack of environmentally friendly initiatives. The natural capital

containing environmental conditions is the highest challenge for sustainable development (Ibrahim & Sapanli, 2024). It is still very likely that this condition can be improved to be very sustainable if strategic steps focusing on improving environmental and economic aspects while maintaining social sustainability can be taken.

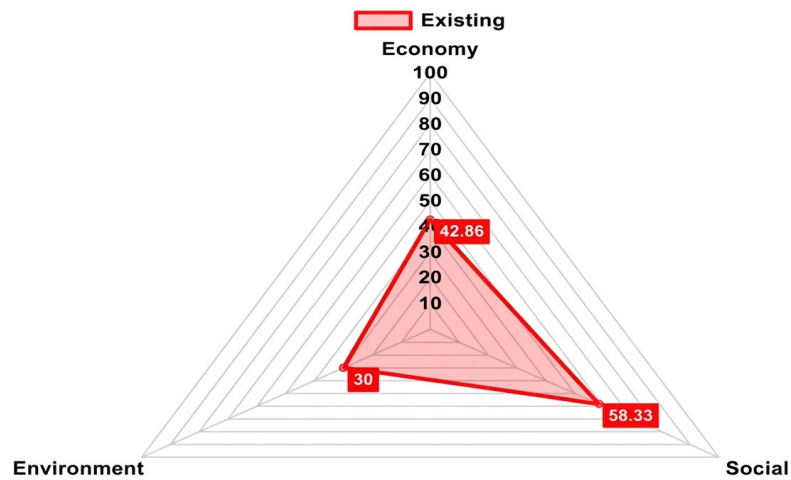


Figure 5. Sustainability status of the aggregate aspect

Table 3. Sustainability score

No.	Aspect	Existing
1	Economic	42.86
2	Social	58.33
3	Environmental	30
	Average	43.73
	Status sustainability	Low sustainable

3.2.2 Leverage factor

A. Leverage factor in the economic aspect

As shown in Figure 6, sensitivity analysis shows that several main factors contribute to the sustainability of the system in the economic aspect. The factor with the highest maximum sensitivity value is technology development, with a maximum sensitivity value of 0.5 and a sensitivity value of 1. This shows that technology development is a driving factor or leverage factor in the economic aspect. This means that this factor will affect the sustainability of the economic aspect in the future.



Figure 6. Leverage score of the economic aspect

Other factors such as the production/fishing capacity of mangrove crabs in technology development have a maximum sensitivity value of 0.5 and a sensitivity value of 0. This indicates that production capacity is also an important factor that needs to be considered, although it has not yet had a direct impact. In addition, factors such as sales target and labor income each have a maximum sensitivity of 0.25 and a sensitivity value of 1. This indicates that these factors have the potential to be maximized because they contribute to improving the sustainability of the economic aspect even though they are not the driving factors. Likewise, the market access factor has a maximum sensitivity value of 0.5 and a sensitivity value of 0.5.

The net profit factor has a maximum sensitivity value of 0.25 and a sensitivity value of 0. This suggests that this factor has relatively little potential in influencing sustainability. The product quality has a maximum sensitivity value of 0.17 and a sensitivity value of 0.5, which shows that the effect is not high enough, but it is still important for sustainability improvement. Overall, these results suggest that although all factors have the potential to contribute to sustainability, existing conditions suggest that their sensitivity is not yet optimal. From these sensitivity values, it can be observed what priorities need to be made to improve sustainability. Table 4 shows the priorities.

Table 4. Sustainability development priorities in the economic aspect

Name of Factor	Priorities
Conducting technology development (X_7)	1
Mangrove crab sales target (X_2)	2
Labor income (X_4)	3
Access to the market (X_6)	4
Product quality (X_5)	5
Net profit (X_1)	6
Mangrove crab production/catch capacity (X_3)	6

A very sensitive factor in economic sustainability is productivity enhancement through technology development. This finding contrasts with the results of the multi-aspect sustainability analysis conducted by Lindawati et al. (2024), who considered technology development as a relatively insensitive factor within the rice marketing supply chain sustainability system. This indicates fundamental differences between the terrestrial agricultural and fisheries sectors in terms of production characteristics, operational efficiency, and technological needs.

B. Leverage factor in the social aspect

Based on the results of the sensitivity analysis in Figure 7, several social factors have significant potential in influencing the sustainability of the system. The factor with the highest maximum sensitivity value and real contribution is the use of work safety tools/costumes, which have a maximum sensitivity of 1 and a sensitivity value of 1. This shows that this factor is a driving factor for social sustainability that must be prioritized. In addition, workforce training, such as training in capture, processing, and marketing, also contributes with a maximum sensitivity value of 0.5 and a sensitivity value of 1. This indicates that upskilling the workforce through training has a significant direct impact on sustainability.

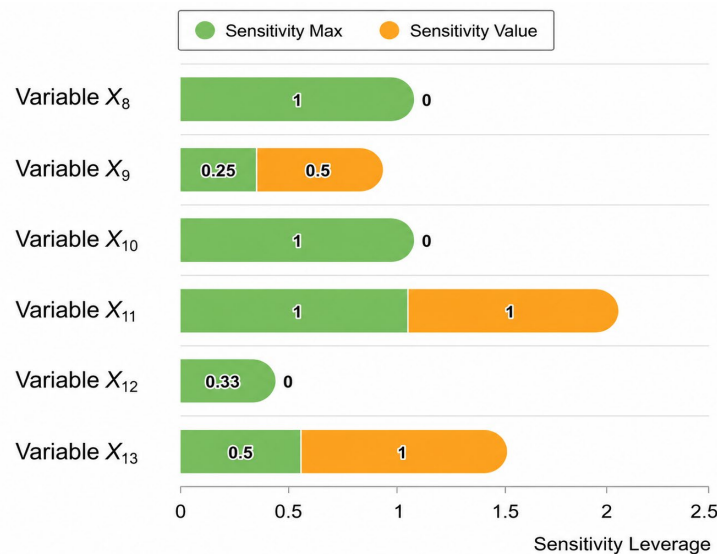


Figure 7. Leverage score of the social aspect

Meanwhile, the labor creativity factor has a maximum sensitivity value of 0.25 and a sensitivity value of 0.5, which shows that labor creativity has an impact, although the potential contribution is still relatively moderate compared to other factors. On the other hand, factors such as the absorption of local labor, ease of access to health facilities, and workplace conditions have maximum sensitivity values of 1, 1, and 0.33, respectively, but with a sensitivity value of 0. This suggests that while these three factors have a potentially significant impact on sustainability, they cannot yet be a driver of future sustainability or have low priority. Overall, the driving factor from the social aspect is the use of work safety tools/costumes with a combination of a maximum sensitivity and sensitivity value of 2. With these results, priority factors can be designed and prioritized to improve the sustainability of the supply chain, as shown in Table 5.

Table 5. Sustainability development priorities in the social aspect

Name of Factor	Priorities
Using work safety tools/costumes (X_{11})	1
Workforce training (X_{13})	2
Workforce creativity (X_9)	3
Labor absorption (X_8)	4
Easy access to health (X_{10})	4
Workplace conditions (X_{12})	4

The supply chain in the marketing of the fisheries sector such as mangrove crabs is still a less sensitive factor in affecting the sustainability level of a supply chain. In contrast to the research conducted by Ibrahim & Sapanli (2024), who considered working conditions as the most sensitive factor in affecting the supply chain sustainability. Meanwhile, the use of costumes and safety equipment is the most sensitive factor in the mangrove crab sector. However, its priority is only at the middle level in the study by Ibrahim & Sapanli (2024). This indicates that sustainability drivers differ across sectors. The fisheries supply chain, particularly in mangrove crab marketing, operates under working environments and operational conditions that differ substantially from those of terrestrial agricultural supply chains.

C. Leverage factor in the environmental aspect

The results of the analysis in Figure 8 show that several environmental factors significantly contribute to the sustainability of the system. Three main factors are the driving factors, namely the reduction of the amount of energy used (gasoline and electricity), the use of environmentally friendly materials, and the ownership of environmentally friendly certificates from credible institutions. Those factors have a maximum sensitivity value of 1 and a sensitivity value of 1. This means that these factors need to be prioritized to increase sustainability in the environmental aspect.

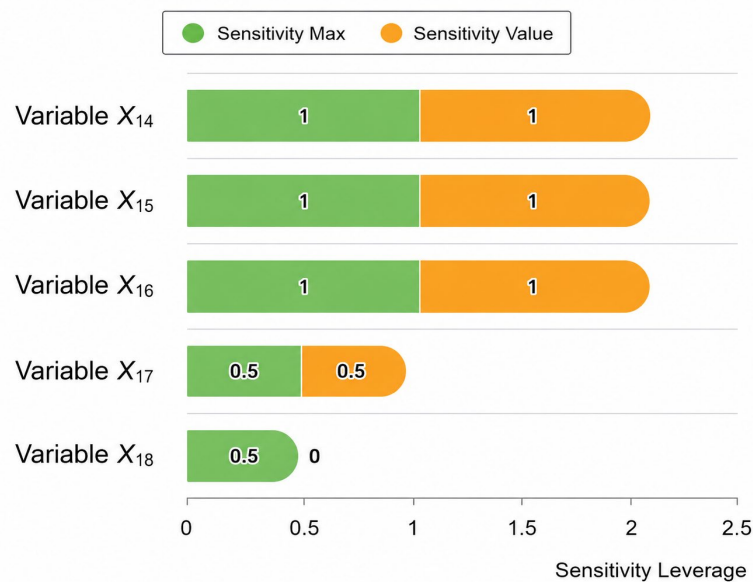


Figure 8. Leverage score of the environmental aspect

Waste management, such as the use of crab shells, has a maximum sensitivity value of 0.5 and a sensitivity value of 0.5, which shows a moderate contribution to sustainability. While its contribution is already visible, this

factor still has room for improvement to maximize its impact on supply chain sustainability. Meanwhile, the use of business inputs from local suppliers has a maximum sensitivity value of 0.5 but a sensitivity value of 0. This suggests that although this factor has the potential to support sustainability, its current contribution is still not optimal and needs to receive special attention in strategic planning. Overall, the results of this analysis emphasize the importance of maintaining and strengthening factors with a real impact, such as energy reduction and the use of environmentally friendly materials. In addition, the formalization of waste management and the use of local business inputs are also priorities in increasing sustainability in the environmental aspect. The order of priorities in improving the sustainability of the environmental aspect is presented in the following table.

Table 6. Sustainability development priorities in the environmental aspect

Name of Factor	Priorities
Reduction in energy (X_{14})	1
Use of environmentally friendly materials (X_{15})	1
Have an eco-friendly certificate (X_{16})	1
Waste management (X_{17})	2
Use of business inputs from local suppliers (X_{18})	3

Reduction in energy, which is not environmentally friendly, is still the most sensitive factor compared to other factors based on Table 6. In fact, the research results from Ibrahim & Sapanli (2024) show that waste management is the most sensitive thing. This indicates that the use of waste or waste produced is relatively better managed in the mangrove crab sector compared to other capture fishery sectors.

4. Conclusion

Sustainable supply chain management analysis using the multi-aspect sustainability analysis method can be a policy-making reference, because it not only assesses the current sustainability status of a supply chain but also identifies priority areas for intervention and the potential direction of future sustainability performance. The results of this study show that the sustainability of the mangrove crab supply chain on Bangka Island is still low or in the low sustainability category. The analysis scores of the three aspects used are still diverse. The economic aspect has a score of around 42.86 in the low sustainable category, the social aspect has a score of 58.33 in the sustainable category, and the environmental aspect has a score of 30 in the low sustainable category. The average score is an aggregate score of 43.73, which is in the low sustainable category. However, of all the aspects, almost all of them have a good score related to the value of possible future performance. This means that it is likely that sustainability will be possible in the future.

5. Recommendation

Based on the results of the analysis, the value of the mangrove crab supply chain sustainability index on Bangka Island is still in the relatively low category. Therefore, improvement efforts in various aspects that affect sustainability should be made. Increasing the sustainability level is not only the responsibility of the government but also requires the involvement of academic and business actors in the mangrove crab supply chain. The government can be engaged in the preparation of regulations, coaching programs, and the provision of facility support that encourages sustainable business practices. Academics can contribute through research, technology development, and mentoring and knowledge transfer activities to business actors. Meanwhile, business actors have an important role in implementing sustainability practices in the fishing, cultivation, processing, distribution, and marketing activities of mangrove crabs.

These efforts to improve sustainability can refer to the results of the leverage factor analysis presented in Table 4, Table 5, and Table 6. These leverage factors represent the attributes that have the greatest influence on the improvement of the sustainability index and at the same time form the basis for the formulation of policy priorities and development strategies. These priorities are set based on attributes without full implementation, and more intensive attention and intervention are required. Thus, it is expected that the synergy between the government, academics, and business actors in improving priority attributes can improve the sustainability performance of the mangrove crab supply chain and make it more effective, targeted, and sustainable.

Author Contributions

Conceptualization, K.T.; software, S.K.; formal analysis, A.M.R.; investigation, A.M.R.; writing—original draft preparation, A.M.R.; writing—review and editing, A.M.R.; supervision, S.K. and N.; project administration, S.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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